

WAGNER LEITMOTIF PATTERN RECOGNITION AND MATCHING IN FILM COMPOSER SCORES

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ABSTRACT

We investigate whether Richard Wagner’s leitmotifs from *Der Ring des Nibelungen* leave traceable melodic patterns in scores by Ennio Morricone, Erich Wolfgang Korngold, Howard Shore, John Williams, and Max Steiner. We present a reproducible symbolic-music pipeline implemented with `music21` for transposition-invariant pattern search using three complementary strategies: exact interval matching, Levenshtein-based approximate matching, and melodic contour matching. Ten canonical leitmotifs are used as references and evaluated on 70 MusicXML film scores. A combined score (interval 50%, contour 30%, rhythm 20%) ranks candidate matches and yields 925 matches above threshold. Results show cross-composer distributions consistent with established musicological expectations and position computational retrieval as a transparent companion to expert score analysis.

1. INTRODUCTION

Richard Wagner’s use of leitmotifs transformed musical narrative practice in opera and influenced later dramatic scoring traditions [1, 2]. Film-music scholarship has documented the migration of Wagnerian thematic techniques into classical Hollywood and beyond [3–5]. Korngold and Steiner are central historical links in this transfer [6, 7], while Shore’s *The Lord of the Rings* scores offer a modern large-scale leitmotivic system [8], standing alongside John Williams’s revival of the practice through his developed thematic architecture for the *Star Wars* saga [9].

Computationally, leitmotif tracing is challenging because motifs appear with transposition, rhythmic variation, and ornamentation. These transformations complicate direct pattern matching and require representations that capture structural similarity while remaining invariant to surface-level changes. Prior Wagner-focused MIR work has centered on the audio domain, using machine-learning approaches for leitmotif classification and activity detection in opera recordings [10, 11].

In contrast, the question of whether Wagner-related melodic patterns can be detected in later repertoires has re-

ceived less computational attention, despite longstanding musicological claims that Wagnerian techniques shaped the development of film scoring. Many film composers consciously adopted thematic transformation, recurring character motifs, and large-scale motivic networks reminiscent of Wagner’s dramaturgical approach. If such influence extends beyond compositional philosophy to the level of melodic structure, then traces of Wagnerian patterns might appear, intentionally or stylistically in later scores.

This raises an empirical question: can computational analysis reveal measurable melodic similarities between Wagner’s leitmotifs and themes found in film music? Detecting such correspondences does not imply direct quotation or deliberate borrowing; rather, it can reveal structural affinities that emerge through shared stylistic conventions. By analyzing symbolic scores rather than audio recordings, we can examine melodic patterns directly and test whether motifs from Wagner’s *Ring* cycle appear in transformed or approximate forms within the thematic material of film composers. Unlike previous motif-detection systems designed for within-work analysis, our approach targets cross-corpus retrieval, enabling the systematic comparison of Wagnerian motifs with thematic material across multiple film composers.

Our work investigates whether melodic patterns derived from Wagner’s leitmotifs can be detected in film-composer scores through symbolic cross-corpus analysis. To address this question, we develop a computational pipeline for leitmotif retrieval in symbolic music data. The method searches for transposition-invariant melodic patterns using complementary similarity measures that capture different aspects of melodic structure.

Our contributions are threefold. First, we present an open symbolic pipeline for transposition-invariant leitmotif retrieval implemented with `music21` [12]. Second, we introduce a unified scoring framework that combines exact interval matching, approximate matching based on Levenshtein distance, and melodic contour similarity. Third, we provide an empirical analysis of Wagner-related melodic patterns across a corpus of 70 MusicXML film scores by five composers: Morricone, Korngold, Shore, Williams, and Steiner.

The implementation and dataset are available at: <https://github.com/federico-de-lellis/wagner-leitmotif-pattern-film-scores>.



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2. BACKGROUND AND RELATED WORK

2.1 Leitmotif and film scoring

Leitmotifs are recurring musical ideas associated with characters, objects, places, or narrative states [1]. In Wagnerian opera, these motifs function not merely as thematic labels but as dynamic musical elements that evolve throughout the dramatic narrative. Motifs may appear in varied forms through transposition, rhythmic modification, melodic ornamentation, or partial fragmentation, allowing them to adapt to changing dramatic contexts while retaining recognisable structural features.

Wagner’s *Der Ring des Nibelungen* cycle represents the most extensively developed example of a large-scale leit-motivic system, containing hundreds of recurring motifs that interact and transform across the four operas. Modern cataloguing efforts and computational studies have produced curated motif inventories and annotated datasets that make systematic analysis possible [10, 13]. These resources provide practical subsets of motifs that can serve as reference patterns for computational study.

Film-music scholarship has long noted the continuity between Wagnerian dramaturgy and the thematic practices of classical Hollywood scoring [2, 3]. Early film composers such as Erich Wolfgang Korngold and Max Steiner adopted the technique of associating recurring musical themes with narrative elements, using them to support character identification, emotional cues, and large-scale narrative coherence. Later composers, like John Williams and Howard Shore expanded these approaches in modern film scores, often constructing extensive thematic networks that function in ways comparable to Wagnerian leit-motivic systems.

2.2 Computational pattern discovery

Pattern-discovery research in music information retrieval highlights several core challenges, including the choice of representation, the handling of variation through approximate matching, and the filtering of retrieved candidates [14, 15]. Melodic patterns may appear in transformed forms through transposition, rhythmic alteration, or ornamental variation, requiring representations that capture structural similarity while remaining robust to such transformations.

Interval-based representations are widely used because they provide natural transposition invariance and allow melodies to be compared independently of absolute pitch. Contour abstractions further reduce melodic information to directional movement, capturing higher-level melodic shape while tolerating small pitch differences [16]. These representations are often combined with string-based comparison techniques to detect recurring melodic fragments within or across musical works.

Approximate string matching methods, including edit-distance measures such as the Levenshtein distance, have been shown to be effective for retrieving melodic patterns under variation [17]. Such approaches enable flexible comparison between motif templates and candidate

melodic segments while allowing limited substitutions, insertions, or deletions. In this study, we adopt these principles to implement a transposition-invariant symbolic retrieval pipeline that combines interval, contour, and rhythmic similarity measures.

3. DATASET

We use ten Wagner leitmotifs (following the subset used in prior Wagner MIR studies [10]) and 70 film-composer MusicXML scores grouped by composer: Morricone, Korngold, Shore, Williams, and Steiner. Scores are processed symbolically; the top melodic part is extracted for this version of the pipeline.

4. METHODOLOGY

4.1 Representations

For each score and leitmotif, we compute four representations: MIDI pitches, chromatic intervals, contour tokens (-1 , 0 , and $+1$), and durations. Interval sequences provide transposition invariance. Contour features capture melodic shape-level similarity [14, 16].

4.2 Matching stages

- **Exact matching:** sliding-window equality on interval and rhythmic sequences.
- **Approximate matching:** Levenshtein distance with bounded edits for near-matches [17].
- **Contour matching:** proportion of equal contour tokens per window.

4.3 Combined score and filtering

Candidates are ranked by

$$\text{score} = 0.50 s_{\text{int}} + 0.30 s_{\text{cont}} + 0.20 s_{\text{rhythm}}. \quad (1)$$

Matches below the minimum threshold are removed; overlapping windows are deduplicated by retaining higher-scoring candidates.

5. RESULTS

The system returns 925 matches above threshold, with strong matches dominating and only a small near-exact subset. Matches are unevenly distributed across composers (Table 2), with Williams highest in raw count and Shore lowest. Qualitatively, many top-scoring matches involve short motifs (especially Mime), while longer lyrical motifs occur less frequently. This aligns with expected combinatorial behavior of short vs. long melodic patterns.

5.1 Manual inspection of matches

To better understand the musical relevance of the automatically retrieved candidates, we conducted a manual inspection of ten randomly selected matches. Each candidate

Table 1. Overview of the Wagner leitmotifs used in this study.

#	Name (English)	German / Common Name	ID (Krause et al. 2020)	Score
1	Forest Murmurs	Waldweben	L-Wa	
2	Horn	Horn	L-Ho	
3	Mime	Mime	L-Mi	
4	Nibelungs	Nibelungen	L-Ni	
5	Nibelungs' Hate	Nibelungenhass	L-NH	
6	Ride	Ritt	L-RT	
7	Ring	Ring	L-Ri	
8	Siblings' Love	Geschwisterliebe	L-Ge	
9	Swirling Blaze	Waberlohe	L-WL	
10	Sword	Schwert	L-Sc	

Table 2. Leitmotif $\times$ composer match counts (above threshold).

Leitmotif	Morricone	Korngold	Shore	Williams	Steiner
ForestMurmurs	1	0	0	5	0
Horn	12	5	4	22	10
Mime	62	49	32	78	56
NibelungsHate	5	13	4	21	8
Nibelungs	44	20	20	35	36
Ride	4	22	1	21	11
Ring	8	9	12	22	23
SiblingsLove	19	6	2	10	7
SwirlingBlaze	48	29	15	46	48
Sword	3	13	0	1	3
Total	206	166	90	261	202

was evaluated by comparing the extracted melodic segment with the corresponding Wagner motif in terms of interval structure, rhythmic profile, and musical context.

The analysis revealed that several matches correspond to short melodic fragments that share interval patterns with Wagner motifs but do not necessarily function as independent motifs within the target scores. For example, some high-scoring detections of the *Mime* motif occur at phrase transitions or as brief melodic gestures rather than structurally meaningful thematic material. This behaviour is expected given the short length of the motif (four notes), which increases the likelihood of coincidental interval matches.

Other cases present more musically plausible correspondences. A match of the *Nibelungs* motif in Howard Shore’s *The Shire* theme exhibits strong contour and rhythmic similarity, suggesting a meaningful structural resemblance within the melodic phrase. Similarly, a passage from Korngold’s *The Sea Hawk* (within the so-called “Spanish Theme”) displays interval and contour patterns comparable to the *Nibelungs* motif.

Some detections highlight limitations of purely melodic

retrieval. For instance, one candidate involving the *Sword* motif occurs in Mendelssohn’s *Wedding March*, a work composed decades before Wagner’s *Ring*. This case illustrates how the method may surface structurally similar melodic fragments even when historical influence is impossible.

Overall, the manual evaluation suggests that the pipeline is effective at identifying structurally similar melodic fragments but that short motifs and rhythmic variation can produce false positives. These results support the interpretation of the system as a candidate-retrieval tool that requires musicological validation rather than a definitive detector of leitmotivic influence.

The ten randomly selected matches that were inspected include near-exact detections in Shore and recurring high-confidence detections in Steiner and Korngold scores, consistent with documented Wagnerian lineage in film scoring [6–8].

6. DISCUSSION

The results suggest that the proposed pipeline can successfully surface melodic fragments that exhibit structural similarity to Wagnerian leitmotifs across a diverse corpus of film scores. In particular, the combination of interval-based matching, contour similarity, and approximate string matching enables the system to retrieve candidates that preserve different aspects of melodic structure under transformation. The manual inspection of detected matches (Table 3) indicates that some retrieved passages correspond to musically meaningful similarities within broader melodic phrases, while others reflect shorter coincidental interval patterns.

A key strength of the approach lies in its transparency and reproducibility. Because the method operates entirely on symbolic representations and uses clearly de-

Table 3. Manual evaluation of 10 detected matches.

Rank	Leitmotif	Composer	Work	Score	Auto Class	Manual Evaluation
1	Mime	Max Steiner	The Caine Mutiny March	0.933	Strong	Not a clear match; phrase ending fragment
2	Nibelungs	Howard Shore	The Shire	0.912	Near-exact	Partial melodic resemblance within phrase
3	Nibelungs	E.W. Korngold	The Sea Hawk (Spanish Theme)	0.838	Near-exact	Theme shows similarity to motif
4	Mime	John Williams	Schindler's List (Main Theme)	0.833	Strong	Very short coincidence; unlikely motif usage
5	Mime	Ennio Morricone	A Fistful of Dollars	0.833	Strong	Very short coincidence; unlikely motif usage
6	Sword	E.W. Korngold	Wedding March (Mendelssohn)	0.833	Approx.	Earlier work; similarity present but historically unrelated
7	Siblings' Love	Ennio Morricone	Once Upon a Time in the West	0.833	Strong	Some melodic and expressive similarity
8	Mime	Howard Shore	Over the Misty Mountains Cold	0.833	Strong	Similar melodic gesture within phrase
9	Nibelungs	John Williams	E.T. Flying Theme	0.812	Near-exact	Some melodic resemblance within theme
10	Swirling Blaze	Max Steiner	Gone With the Wind Suite	0.800	Approx.	Not a match; rhythmic structure differs

finer similarity measures, each detected match can be inspected and interpreted directly in score form. This contrasts with many audio-domain leitmotif detection systems [10, 11, 18], which rely on machine-learning models that may be less interpretable in terms of specific musical structures. While symbolic analysis sacrifices timbral and harmonic information available in audio recordings, it provides a direct representation of melodic content and enables straightforward cross-score comparison.

At the same time, several limitations affect the interpretation of the results. The current pipeline analyzes only the top melodic voice of each score, potentially overlooking relevant material in inner voices or orchestral textures. In addition, rhythmic similarity is treated in a simplified manner and approximate matching relies on uniform edit costs, which may not fully reflect perceptual or stylistic differences between melodic variants. These constraints mean that the system should primarily be understood as a candidate-retrieval tool rather than a definitive detector of leitmotivic relationships.

Importantly, the presence of melodic similarities should not be interpreted as evidence of intentional quotation or direct borrowing. Many film composers working within the late-Romantic orchestral tradition share common stylistic vocabularies derived from nineteenth-century harmonic and melodic practices. As a result, structurally similar melodic fragments may arise independently or through unconscious stylistic influence rather than deliberate reference to specific Wagner motifs. The matches identified by our system should therefore be interpreted as indicators of structural affinity rather than definitive proof of leitmotivic borrowing.

7. CONCLUSION AND FUTURE WORK

We presented a symbolic pipeline for Wagner leitmotif matching in film-composer scores. The unified method combines exact, approximate, and contour matching, and it identified 925 above-threshold matches across 70 scores.

Future work includes multi-voice analysis, rhythm-aware matching, musically weighted edit distances, and statistical null-model testing. Integration with annotated resources such as the Wagner Ring Dataset can further support evaluation and interpretation [13].

8. AI USAGE STATEMENT

AI-assisted editing support was used to refine technical writing and formatting.

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